RADIANT PHARMACEUTICALS LIMITED

ICT Wing

(Internship Daily Report)

INTERN PROFILE	DETAILS
Intern Name	Ibrahim Hasan
Student ID	2110316
University	Independent University, Bangladesh
Department	Management Information System (MIS)
Supervisor	Ayan Chowdhury (Sr. Officer, ICT)

(Daily Progress Report)

Day #	Date #
Day-6	24-05-2026

(Daily Report)

Task Assigned:

- Build a Python-based data pipeline to read, clean, sort, and bulk-insert vehicle sales data (up to 1500 rows) into a MySQL database.
- Ensure data cleaning filters out all missing/placeholder values and database insertion aligns with the existing 17-column layout (with saledate as VARCHAR).
- Implement a separate Python data analysis script using Pandas to read, clean, process, and analyze the full 558,837-row vehicle sales dataset and output key business insights.

Learning Outcome:

- Learned to stream large CSV files dynamically using Python's standard csv.DictReader to prevent memory overflow.
- Understood database schema alignment: matched insertion logic to an existing VARCHAR-based saledate schema, sorting chronologically by helper datetime objects in memory and preserving order via MySQL auto-increment IDs.
- Optimized data analysis runtime performance in Pandas: replaced slow regex-based cell replaces with hash-map static value replacements and vectorized string split/concatenations, reducing execution time from minutes to under 5 seconds.
- Handled standard output Unicode encoding issues on Windows consoles by configuring table outputs to utilize ASCII-only psql-style rendering.

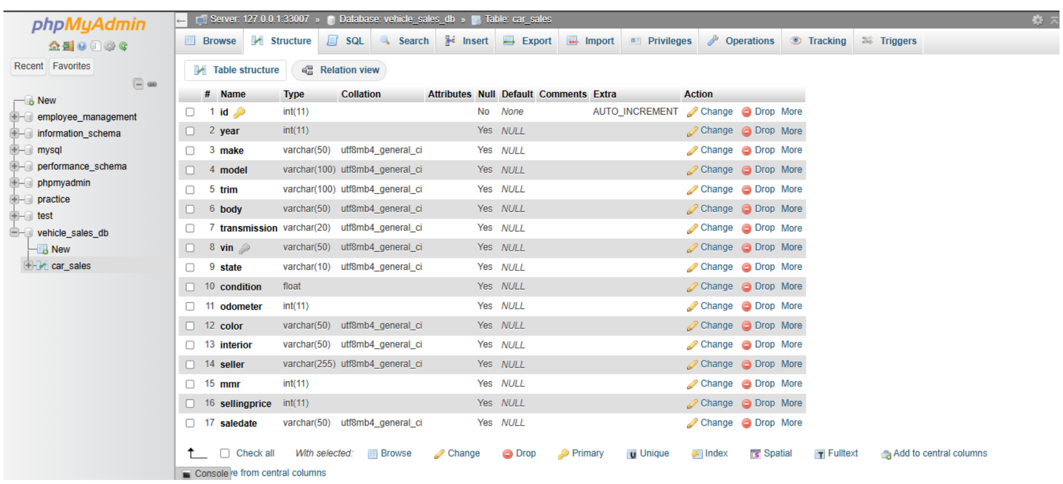

Figure 1: Vehicle Sales MySQL Database.

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====================================================
                BUSINESS INSIGHTS
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1. Top 5 Vehicle Makes by Sales Volume:
    Make  Sales Volume
Infiniti           215
  Nissan           186
 Hyundai           170
     Bmw           167
    Ford           100

2. Top 5 Most Profitable Vehicle Makes (Average Profit/Loss):
   Make Avg Profit/Loss ($)
Ferrari           $5,000.00
  Smart           $1,275.00
  Dodge           $1,089.58
  Mazda             $943.75
Porsche             $767.86

3. Monthly Sales Volume and Profit Trend:
 Year    Month  Sales_Count  Total_Revenue Total_Profit
 2014  January            1     $19,000.00    $6,950.00
 2014 December         1321 $26,432,603.00 $-100,247.00
 2015  January          154  $3,529,500.00 $-247,100.00
 2015 February           22    $555,450.00  $-29,675.00
 2015      May            1     $15,000.00   $-2,650.00
 2015     July            1      $7,200.00  $-17,100.00

4. Transmission Type Comparison:
Transmission  Sales Volume Avg Selling Price Avg Profit/Loss
   automatic          1459        $20,470.36         $-265.40
      manual            41        $16,890.24          $-63.41
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Figure 2: Pandas Data Analysis & Business Insights (analysis.py Output)

Work Completed:

- Created .env configuration files for local database port 33007 connection details.
- Developed and successfully executed pipeline.py to load, sort, and insert 1500 cleaned records.
- Created and ran verify.py to run data quality and chronological sorting checks against MySQL.
- Created and ran analysis/analysis.py to load, clean, and output business insights from 558,837 rows.
- Implemented query_runner.py to execute arbitrary queries on the database interactively.

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Problems Faced:

- Port connection refusal on the default MySQL port 3306.
- Severe processing freeze/delay during Pandas string placeholder cleanups on the large 558k dataset.
- UnicodeEncodeError when displaying table format boundaries to the default Windows shell.

Solution Attempted:

- Used PowerShell net stat check to discover the MySQL service binds to port 33007, and updated database configurations.
- Refactored data cleaning logic to strip whitespace first and apply direct list replacement, avoiding regex compilers.
- Updated output styling to use ASCII characters (tablefmt='psql') to render grids safely in all shell encodings.